KSHITIJ SHARMA

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Objective

Aspiring bioinformatics researcher with a strong interest in AI-driven biomarker discovery and molecular diagnostics. Passionate about integrating computational biology, OMICS data analysis, and molecular techniques to develop early diagnostic solutions for cancer and genetic disorders.

Experience

- **Helix Biogenesis Pvt. Ltd.** 08/11/25 - 29/11/25
  Intern
  ●Performed bacterial isolation, staining, and biochemical identification
  ●Conducted DNA isolation, PCR, and agarose gel electrophoresis
  ●Gained hands-on experience in GLP, buffer preparation, and antimicrobial testing (AST, MDR)

- **AIIMS Raipur (via Ethical Edufabrica)** 24/03/26 -
  Intern
  ●Worked on CRISPR-Cas and gene editing (in-silico) alongwith research project.
  ●Performed primer designing, gRNA designing, and homology modeling
  ●Used tools like CHOPCHOP, protein databases, and bioinformatics platforms

Education

Course / Degree	School / University	Grade / Score	Year
M.Sc. Biotechnology 2nd Sem(Pursuing)	INSTITUTE OF INFORMATION & MANAGEMENT TECHNOLOGY ALIGARH	7.8 CGPA	2025-2027
B.Sc. (Botany,Chemistry, Zoology)	Mahatma Gandhi Kashi Vidyapith,Varanasi	66.33%	2022
HSC	Shri H L V Inter College, Chunar, Mirzapur	73.4%	2019
SSC	Scholars Academy, Varanasi	72.2%	2017

Skills

- Bioinformatics: UniProt, ExPASy, SOPMA, AlphaFold, CHOPCHOP
- Molecular Biology: PCR, Gel Electrophoresis, DNA Isolation
- Programming: Python (Basic)
- Data Analysis: Protein structure prediction, sequence analysis
- AI/ML: Supervised learning (Logistic Regression, Random Forest), basic data modeling
- Others: Experimental design, microbiology techniques

Projects

- **Comparative In-Silico Evaluation of MUC1, MMP-9 and TIMP-1 as Potential Early Diagnostic Biomarkers in Pancreatic Cancer (Under publishing by Journal of Proteins& Protenomics,Springer)**
  Conducted computational analysis of cancer-associated proteins for early diagnosis.Performed sequence retrieval, physicochemical characterization, and structure prediction .Utilized UniProt, ExPASy ProtParam, SOPMA, and AlphaFold.Identified MUC1 as a potential early-stage diagnostic biomarker.Demonstrated integration of bioinformatics tools in translational cancer research
- **AI-Driven Biomarker Prediction Using Protein Sequence and Physicochemical Feature Analysis**
  Developed a basic machine learning model to analyze protein sequence-derived features for predicting potential disease biomarkers

Extracted physicochemical properties using bioinformatics tools (ExPASy ProtParam, UniProt).Applied supervised learning algorithms (e.g., Logistic Regression / Random Forest) for classification.Performed data preprocessing, feature selection, and model evaluation.Demonstrated the potential of AI in biomarker discovery and early disease diagnosis

RESEARCH INTERESTS

- ●Cancer biomarker discovery
  ●AI in bioinformatics
  ●OMICS data analysis
  ●Genetic disorders
  ● Computational genomics